

Mathematical modelling and sensitivity analysis of solar photovoltaic panel integrated with phase change material

Tao Ma*, Jiaxin Zhao, Zhenpeng Li

School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China

HIGHLIGHTS

- Enhanced conductivity method is adopted to incorporate PCM convective effect into 1-D thermal resistance model.
- An improved thermal resistance model, offering a good compromise between accuracy and simplicity, is developed.
- Neglecting PCM convective effect or radiative heat transfer in numerical simulation may result in significant errors.
- A sensitivity analysis suggests that the optimal PCM melting temperature is about 5 °C higher than ambient.

ARTICLE INFO

Keywords:

Solar photovoltaic (PV)
Phase change materials (PCM)
PV-PCM
Thermal regulation
Thermal resistance model
Enhanced conductivity method

ABSTRACT

It is reported that every degree rise in photovoltaic (PV) temperature could lead to a decrease in electricity output by 0.4–0.65%. Phase change material (PCM), which could absorb great amount of heat without raising the temperature of itself, is employed in this study to control PV module temperature and increase power generation. This kind of integrated system is the so-called PV-PCM system. In recent years, some work has already been conducted in using PCM for PV panel thermal regulation both numerically and experimentally, while some issues are still unsolved or unclear, for example, limited number of cases in simulation, difficulties in modelling PCM convective effect, the impact and uncertainty resulting from some common assumptions in numerical simulation. To examine these issues, an improved thermal resistance model through applying enhanced conductivity method is developed to incorporate PCM convective effect in 1-D model, offering a good compromise between accuracy and simplicity. The numerical simulation result illustrates that neglecting PCM convective and radiative heat transfer will cause significant errors. Finally, based on simulation of over 300 cases, the two-variable analysis demonstrates that every 100 W/m² increase in solar radiation can lead to about 5 °C increase in peak temperature, and an optimal performance can be achieved when the melting temperature of PCM is slightly higher, such as 5 °C, than the ambient temperature.

1. Introduction

Nowadays, solar energy harnessed by photovoltaic (PV) panels is regarded as one of the most promising energy sources to deal with world energy crisis and global warming [1]. For the purpose to generate more electricity from the same amount of solar energy, scientists relentlessly pursue higher and higher PV conversion efficiency [2]. However, not only the characteristics of PV panel itself determine the PV conversion efficiency, the ambient environment also has impact on it [3]. While under laboratory conditions, the efficiency could reach to over 30% [4], only about 5–20% solar energy can be converted to electricity under real conditions [5]. Various environmental factors, e.g., the intensity of solar radiation and the dust accumulation on the

PV surface, can affect the efficiency. Moreover, the high PV temperature has negative impact, $-0.4\%/^{\circ}\text{C}$ to $-0.65\%/^{\circ}\text{C}$, on PV conversion efficiency, [6], which has been extensively reported in academic circles, for example [7,8]. As the PV temperature could rise up to 80 °C and even higher than 100 °C in desert area, a significant reduction in PV power generation was observed [9].

In this context, researchers have adopted various thermal management techniques to mitigate and control the PV surface temperature [10], e.g., natural/forced air cooling [11], water cooling [12] and heat pipe [13]. Generally, those cooling techniques could be classified into two categories: active cooling and passive cooling [14]. Active system consumes extra electricity to drive cooling water or other coolants to cool the PV panel which could achieve better cooling effect than passive

* Corresponding author.

E-mail address: tao.ma@connect.polyu.hk (T. Ma).

Nomenclature

ρ	density (kg/m ³)
μ	dynamic viscosity of phase change material (N s/m ²)
C_p	specific heat (J/(kg K))
g	gravity acceleration (m/s ²)
Δt	time interval (s)
A	area of PV panel (m ²)
α	absorptivity of PV panel
G_{PV}	solar radiation on PV panel (W/m ²)
η	actual efficiency of PV panel (%)
$\varepsilon_{\text{glass}}$	emissivity of PV surface
F	view factor of PV surface
σ	the Stefan–Boltzmann constant
T_{surface}	surface temperature of PV-PCM system (°C)
T_{sky}	sky temperature (°C)
h_{free}	natural heat convection coefficient (W/(m ² K))
h_{forced}	forced heat convection coefficient (W/(m ² K))
ε_{Al}	emissivity of aluminum
T_{back}	back surface temperature of PV-PCM system (°C)
T_{ground}	ground temperature (°C)
Q	net solar radiation input (W)
q	total heat input (W)

q_{sky}	radiative heat loss to sky (W)
T_{amb}	ambient temperature (°C)
q_{ground}	radiative heat loss to ground (W)
k	thermal conductivity (W/(m K))
L	height of PV panel (m)
Ra	Raleigh number
Pr	Prandtl number
Re	Reynolds number
ρ_s	density of PCM in solid state (kg/m ³)
ρ_l	density of PCM in liquid state (kg/m ³)
c_s	specific heat of PCM in solid state (J/(kg K))
c_l	specific heat of PCM in liquid state (J/(kg K))
T_l	liquidus temperature of phase change material (°C)
T_s	solidus temperature of phase change material (°C)
E	energy absorbed by phase change material (J/kg)
E_0	energy required to rise to the solidus temperature of phase change material (J/kg)
E_l	latent heat capacity of phase change material (J/kg)
P_{out}	output power of PV panel (W)
η_{ref}	reference efficiency of PV panel
β_{ref}	coefficient between PV temperature and efficiency
k_{PCM}	thermal conductivity of PCM layer (W/(m K))

systems but usually costs more. In 1978, Stultz et al. [15] proposed to use phase change material (PCM) to cool the PV panel since PCM has the latent heat capacity which can absorb substantial amount of heat from PV. From that on, some research has been carried out in the field of PCM for PV temperature regulation, that is so-called PV-PCM system. Huang et al. has studied the performance of PV-PCM system comprehensively in earlier years, for example, the 2-D computational fluid dynamics (CFD) model was firstly employed to analyze the micro-level operational variables in the PV-PCM system [16], i.e., temperature, velocity fields and vortex formation, and it was found that the existence of PCM could enhance PV conversion efficiency significantly [17]. After that, more papers were published to investigate different issues, such as impact of different types of PCM, impact of fins attached to the PCM chamber, melting behavior of PCM [18] and comparison between 3-D and 2-D CFD model [19].

At the same time, more in-depth research has been conducted by other scholars to broaden the scope of study in PV-PCM. For example, Hasan et al. [20] performed a techno-economic study about PV-PCM system under two locations with different climates, i.e., Dublin and Vehari. The simulation result illustrates that the peak temperature of PV-only system and PV-PCM system is 63 °C and 42 °C respectively in a Vehari's summer day, and the average efficiency can be improved by about 10%. It also concluded that the PV-PCM system was financially viable in Vehari's hot climate but not in Dublin's mild climate. A year-round numerical study conducted in United Arab Emirates suggests that in extremely hot climate, the PV-PCM system can raise the annual electricity yield by 5.9% [21]. One more study was conducted about the building-integrated photovoltaic panel (BIPV) with PCM, in which the indoor thermal effect of PV-PCM system was examined and compared with PV-only system, revealing that PV-PCM system not only could improve the PV efficiency but meanwhile decrease the heat flux going into indoor environment [22]. Besides, a multi-location study suggests that the BIPV-PCM system successfully reduces the indoor cooling load by 20–30% in Venice, Helsinki and Abu Dhabi, and meanwhile, the peak value of electricity generation is improved by 5–8% [23]. Another study also proves that the PV-PCM system could improve the electricity production by 1–1.5% annually in South Korea [24]. On the contrary, an experimental study from Japs et al. [25] indicates that from July to August in Germany, both the electricity generation and economic profit improvement are mostly negative compared to traditional PV module,

which is mainly attributed to the low thermal conductivity of PCM. However, a recent experimental study of BIPV-PCM system completed in India demonstrates again that the efficiency can be improved by 10% daily compared to the reference BIPV system [26]. Moreover, a global analysis was presented by Smith et al. [27]. It reveals that by choosing the optimal PCM melting temperature, over 6% more electricity generation could be obtained in Mexico and eastern Africa. In many regions, e.g. Central and South America, Africa and Southern Asia, 5% enhancement is possible, while Europe might be the unfavorable region due to only about 2–5% enhancement. Similar to Hasan's indoor study [22], a field study of a ventilated solar roof with PCM suggests that the cooling load is significantly decreased by adopting the PV-PCM roof in East Tennessee, USA [28]. Furthermore, various numerical modelling methods, including CFD model [29], thermal resistance model [30] and finite element model [31], are also introduced to simulate PV-PCM performance under various conditions.

Another field of PV-PCM research is to utilize PCM for other kinds of PV system, e.g., concentrated PV (CPV) system, PV/thermal (PV/T) system and photovoltaic-thermal electric (PV/TE) system. Substantial amount of studies has been conducted on the CPV-PCM system since CPV technology can obtain very high temperature. It is reported that a V-trough CPV installed with phase change material matrix obtains 55% more electricity output than the reference system [32]. Another study demonstrates that CPV-PCM system with 45° tilted angle has optimal performance and uniform temperature distribution on the PV surface [33]. It is believed that CPV integrated with PCM could be a promising and practical application of PV-PCM system in the near future [34]. Besides, some studies attempted to combine the traditional PV/T system with PCM [35]. One numerical study of PV/T-PCM system indicates that 9% increase in electricity output and 20 °C increase in water temperature could be achieved at a high solar radiation and mild ambient temperature day [36]. The PV/TE-PCM system, a relatively novel concept, was investigated by Cui et al. [37], suggesting that it seems not a feasible option under current technologies since the PV module contributes to 98% electricity output while the thermal electric module only accounts for 2%. Moreover, some novel improvement approaches in PCM have been examined, e.g., microencapsulated PCM [38,39] and using pork fat as a cheap source of PCM [40].

Despite that obvious progress has been made by many scholars in this field, several issues are still unclear and have not been identified in

those studies. For instance, numerical studies usually have some assumptions in their mathematical modeling [41], e.g., neglecting radiative heat transfer [42] or neglecting convective heat transfer within PCM [43], whereas the impact of those assumptions on simulation results are poorly discussed in the literature. Besides, to achieve an optimal system configuration, limited cases, usually in the range of 2–20, were compared through changing a single design parameter or boundary condition in numerical [44] and experimental studies [45]. These comparisons are a good start but such investigation cannot achieve a real optimal PV-PCM system configuration for practical implementation. One major reason for these problems is the limitation in mathematical models and numerical methods. As convection occurs in the liquid PCM practically, Navier-Stokes equation is required to be numerically solved. Consequently, the computation time becomes quite long, making it is nearly impossible to conduct substantial number of simulations in one study. To avoid this problem, some researchers assumed that the convective effect can be abandoned and only conductive heat transfer was considered [41,43]. However, such assumption has not been well justified. Fortunately, the research on PCM heat transfer provides an effective solution to them [46]. Based on scaling theory and enhanced conductivity method, the convective effect in PCM can be taken into account through increasing conduction rate [47]. By applying this method, the calculation of Navier-Stokes equation can be avoided. For example, Kahraman et al. applied an enhanced conductivity method in modelling the melting of ice in rectangle enclosure, which shows excellent agreement with numerical solution [48,49]. A similar method was used by Costa et al. to simulate a PCM energy storage system [50]. In recent years, several studies, for example, Refs. [51,52], were carried out to apply the enhanced conductivity method to effectively simulate the movement of melting front of PCM. It is suggested that the computation time can be reduced significantly by applying this method [51].

Therefore, after reviewing previous research in this field and relevant studies in PCM heat transfer, a comprehensive study about PV-PCM system is conducted to overcome the problems as described above. The CFD model and 1-D thermal resistance model are developed for numerical simulation, and a comparison is performed to analyze the advantages and disadvantages between two methods. Based on the comparison, an improved 1-D thermal resistance model through applying the enhanced conductivity method is then developed to highlight the convective heat transfer process taking place within PCM. Moreover, several commonly used assumptions in PV-PCM system modelling are examined in MATLAB to study their impact and uncertainty, aiming to bridge the research gap and offer meaningful

insights for similar numerical studies in the future. Finally, over 300 cases of PV-PCM system with different design parameters and boundary conditions are simulated, and the result is innovatively investigated by a two-variable analysis method to obtain a clear understanding and provide reference for PV-PCM optimization.

2. System configuration and mathematical models

2.1. System configuration

Basically, PV-PCM system has two major parts, i.e. PV panel and PCM chamber. Usually, crystalline silicon PV is used in the system due to its wide utilization and high temperature coefficient. There are several shapes of PCM chamber, e.g., rectangle chamber, semi-circle chamber, triangle chamber and finned chamber in the literature [10], and generally rectangle chamber is selected.

The PV-PCM system in this study is assembled by a PV module and an aluminum chamber filled with PCM. Fig. 1(a) illustrates the three-dimensional graphic of the PV-PCM system and Fig. 1(b) shows the sectional view. Fins in the aluminum chamber is optional and in the 1-D thermal resistance model of this study, they are not applied. Table 1 summarizes the basic properties of PV cell, glass cover, EVA, and chamber used in this study. RT35 from RUBITHERM Technologies is adopted as the PCM in this study since it is commercially available and widely used in the literature [14,25] for the same purpose. Besides, RT35 is non-combustible and non-corrosive which is safe to be filled into the aluminum chamber.

In recent years, the PCM based method for PV thermal regulation has attracted much attention from scholars, and now there are also some demonstration projects in the world. However, the major problem for promoting this technology is the economic issue, it will have a relatively long payback period taking into account the initial cost of PV panels and the additional cost of the PCM. However, if the thermal energy stored in PCMs can be used for the buildings involved such as space heating, ventilation and hot water, and this its economic performance can be improved and show a promising future [10].

2.2. Mathematical models

The mathematical models of the PV-PCM system should be developed first to study the system performance numerically, the models can also be used to investigate how design parameters and boundary conditions impact its performance. Various modelling methods are used in literature, including CFD model [29], thermal resistance model [30]

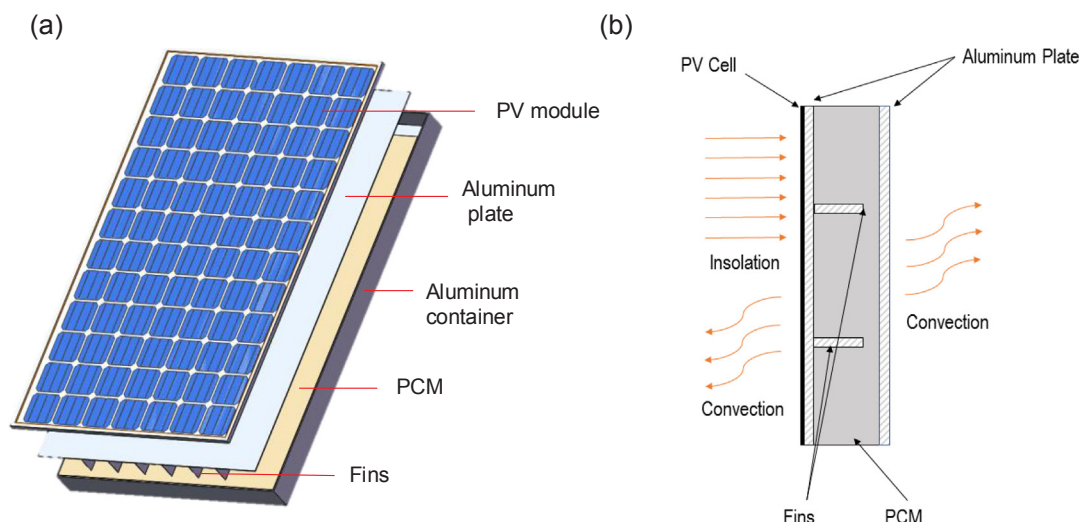


Fig. 1. (a) Three-dimensional view of PV-PCM system; (b) sectional view of the system configuration.

Table 1

The properties of PV module and PCM chamber.

Property	Glass	EVA	Silicon cells	Tedlar	Polyester	RT35 (PCM)	Aluminum
Density (kg/m ³)	2500	935	2330	1500	1390	800	2791
Specific heat (J/(kg K))	750	2500	700	1090	1172	2000	871
Thickness (mm)	3.2	5.0	0.2	0.0375	0.25	20–50	5
Thermal conductivity (W/(m K))	1.04	0.29	150	0.35	0.155	0.2	202.4
Emissivity	0.95	/	/	/	/	/	0.095
Solidus temperature (°C)	/	/	/	/	/	29	/
Liquidus temperature (°C)	/	/	/	/	/	36	/
Latent heat capacity (kJ/kg)	/	/	/	/	/	130	/

and finite element model [31]. In this study, two numerical modelling methods were employed, i.e., CFD simulation and 1-D thermal resistance model. For a typical PV-PCM system, the mathematical model could be generally divided into four main parts, i.e. fluid model, heat transfer model, phase change model and power generation model.

2.2.1. Fluid model

Based on the two-dimensional coordinate system in Fig. 1(b), the momentum equation, i.e., Navier–Stokes equation, is shown below:

$$\rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} = \nabla(\mu \nabla u) - \frac{\partial P}{\partial x} \quad (1)$$

$$\rho \frac{\partial v}{\partial t} + \rho u \frac{\partial v}{\partial x} + \rho v \frac{\partial v}{\partial y} = \nabla(\mu \nabla v) - \frac{\partial P}{\partial y} - \rho g \quad (2)$$

where ρ is the fluid density (kg/m³); u and v is the x-velocity and y-velocity (m/s); g is the gravity acceleration (kg m/s²); μ_l and μ_s is the viscosity of liquid PCM and solid PCM respectively (N s/m²). The same denotation is used for density ρ .

In Eqs. (1) and (2), when PCM is in solid state: $\mu = \mu_s = \infty$, $\rho = \rho_s$; when PCM is in liquid state: $\mu = \mu_l$, $\rho = \rho_l$.

2.2.2. Heat transfer model

Heat transfer in a typical PV-PCM system is depicted in Fig. 1(b). Besides electricity production, solar energy absorbed by the PV panel is mostly converted into heat, which is subsequently transferred to the PCM. It is assumed that there is no heat loss at the surrounding of PV-PCM system as the area is relatively small. The heat conduction within the solid part of the system, i.e., PV module, aluminum plate and the solid PCM, can be expressed as:

$$\rho C_p \frac{\partial T}{\partial t} = k \frac{\partial T}{\partial x} \quad (3)$$

where C_p is the specific heat of each solid part (J/(kg K)) and k is the thermal conductivity of each part (W/(m K)).

At the front and back surface of PV-PCM system, the heat exchange with the ambient environment is dominated by the convective and radiative heat transfer. The energy balance equations are presented by Eqs. (4) and (5):

$$\rho C_p dx \frac{\partial T}{\partial t} = \alpha G_{PV}(1-\eta) - \varepsilon_{\text{glass}} F \sigma (T_{\text{surface}}^4 - T_{\text{sky}}^4) - h(T_{\text{surface}} - T_{\text{amb}}) \quad (4)$$

$$\rho C_p dx \frac{\partial T}{\partial t} = \varepsilon_{\text{Al}} F \sigma (T_{\text{back}}^4 - T_{\text{ground}}^4) + h(T_{\text{back}} - T_{\text{amb}}) \quad (5)$$

where the absorptivity α is a function of absorptive and reflective properties of the encapsulating glass, laminating material and the absorptivity characteristics of the PV cell material, and it is set as 0.9 in this study [44]; G_{PV} is the solar radiation on the PV module (W/m²); $\varepsilon_{\text{glass}}$ and ε_{Al} is the emissivity of the glass and the aluminum plate respectively; T_{surface} , T_{sky} , T_{back} and T_{ground} is temperature of the front surface, the sky, the back surface and the ground respectively (°C); h is the convective heat transfer rate (W/(m² K)), which could be further defined into free convection and forced convection in Eqs. (11) and (12). The net solar radiation casts on PV front surface is given as:

$$Q = \alpha G_{PV} A \quad (6)$$

where A is the surface area of PV-PCM system (m²).

As part of the energy is converted into electricity by photovoltaic effect, the net heat input q is governed by Eq. (7).

$$q = \alpha G_{PV}(1-\eta)A \quad (7)$$

where η is the real efficiency of PV module. The radiative heat transfer in the front and back surface is governed by Eqs. (8)–(10).

$$q_{\text{sky}} = \varepsilon_{\text{glass}} F \sigma (T_{\text{surface}}^4 - T_{\text{sky}}^4) A \quad (8)$$

$$T_{\text{sky}} = 0.0375 T_{\text{amb}}^{1.5} + 0.32 T_{\text{amb}} \quad (9)$$

$$q_r = \varepsilon_{\text{Al}} F \sigma (T_{\text{back}}^4 - T_{\text{ground}}^4) A \quad (10)$$

where the radiative heat transfer with the sky and the ground is represented by q_{sky} and q_{ground} respectively (W). The sky temperature is estimated based on the study of Biwole et al. [53], and the ground temperature is assumed to be equal to the ambient temperature. The convective heat loss at the front and back surface of the PV-PCM system is also considered. The estimation of natural convective rate h_{free} and forced convective rate h_{force} is presented in Eqs. (11) and (12) [54].

$$h_{\text{free}} = \frac{k}{L} \left\{ 0.825 + \frac{0.387 Ra_L^{1/6}}{[1 + (0.492/Pr)^{9/16}]^{8/27}} \right\}^2 \quad (11)$$

$$h_{\text{forced}} = \frac{k}{L} (0.664 Re_L^{1/2} Pr^{1/3}) \quad (12)$$

where L is the height of PV Panel (m); Ra_L is the Raleigh number; Re is the Reynolds number and Pr is the Prandtl number.

2.2.3. Phase change model

For PCM, the heat diffusion equation in liquid state and solid state is given as:

$$\rho C_p \frac{\partial T}{\partial t} + \nabla(-k \nabla T) + \rho C_p \vec{u} \nabla T = 0 \quad (13)$$

As PCM has both latent heat capacity and sensible heat capacity, the temperature during phase change process can be divided into three stages, as shown in Eq. (14).

$$T = \begin{cases} E/c_s + T_0 & E < E_0, T < T_i \\ T_s + ((E-E_0)/E_i)(T_i-T_s) & E_0 < E < E_0 + E_i, T_s < T < T_i \\ T_i + (E-E_0-E_i)/c_L & E \geq E_0 + E_i, T > T_i \end{cases} \quad (14)$$

where E , E_0 and E_i denotes the heat already absorbed by the PCM, the minimum heat required to begin the phase change process and the latent heat capacity of the PCM respectively (J); T , T_s and T_i represents the current PCM temperature, the solidus and liquidus temperature of the PCM respectively (°C). c_s and c_L is specific heat of solid PCM and liquid PCM respectively.

2.2.4. Power generation model

The power generation equation of PV cells is simulated by Eq. (15).

$$P_{\text{out}} = \eta \cdot A \cdot G_{PV} \quad (15)$$

where P_{out} represents the energy output of PV panel (W); η is the real PV conversion efficiency.

The correlation between temperature rise and the decline of PV efficiency is presented in Eq. (16).

$$\eta = \eta_{ref} \cdot \{1 - \beta_{ref} (T_{surface} - T_{ref})\} \quad (16)$$

where the reference PV efficiency, η_{ref} , is 17.1% based on the information provided by the manufacturer; β_{ref} is the temperature coefficient of PV efficiency (0.5%/°C); $T_{surface}$ and T_{ref} represents the real PV temperature and the reference temperature at the standard test condition respectively (°C).

3. CFD simulation and 1-D thermal resistance modelling

After investigating the mathematical models of PV-PCM system above, two numerical methods, i.e., CFD simulation and 1-D thermal resistance modelling, are discussed in this section, and their merits and demerits are analyzed. Finally, the improved 1-D thermal resistance model which requires less computation time is selected for the further study. The analysis below can also provide reference in choosing modelling method.

3.1. CFD simulation

The CFD model was constructed in ANSYS Fluent 17.0 in our previous study [55]. In that study, mesh independence test was conducted, and a comparison with experimental and numerical from other studies were completed, demonstrating a good agreement in patterns of PV temperature profile. A sample mesh of the PV-PCM system constructed in ANSYS is illustrated in Fig. 2. To avoid the mesh dependency program, a grid sensitivity test was conducted and the result is presented in Fig. 3. The very slight temperature difference between the three simulations demonstrates that it is reasonable to set the cell size at 1 mm * 1 mm for the subsequent numerical simulations.

Fig. 4 shows a sample result of CFD modelling. The temperature distribution and flow process of liquid PCM in the system can be observed clearly. Moreover, micro-analysis, e.g., the temperature variation and fluid behavior at each small grid, could also be easily performed by CFD modelling. However, the computation time of CFD simulation is very long, and it often consumes several days to conduct a complete case simulation. That is the major reason why only 2–20 different cases of PV-PCM system were simulated in the literature.

3.2. 1-D thermal resistance modelling

The optimization of a PV-PCM system should compare substantial cases, the CFD simulation seems too complex and not an ideal method for such research. Therefore, another modelling method, i.e., thermal resistance model using finite difference method, is investigated in this study. The schematic diagram of 1-D thermal resistance model is shown in Fig. 5. As indicated, PV-PCM system can be divided into several layers, each layer is a node of the thermal resistance model representing a certain material, e.g., glass, EVA, PV cells, Tedlar and so on. Particularly, the PCM is separated into layers which have a thickness of 1 mm each node to obtain a better accuracy of simulation in PCM. The selection of this node thickness is decided by consideration of avoiding mesh dependence problem and the convenience of the study. Solar radiation, radiative heat transfer, natural air convection and forced air convection are considered at the front surface of the system and the conductive heat transfer is included within the system. At the back of the system, radiative heat transfer and convective transfer with the ambient are also considered. However, it is a challenging work to investigate the convective heat transfer process within PCM because the momentum equations should be considered while the thermal resistance model cannot do that. The solution of this problem is discussed

in the next section.

3.3. A new model for PCM heat transfer rate

To solve the problems mentioned above, an enhanced conductivity model of PCM is proposed in this study. Generally, during the phase change process from solid to liquid, the molten PCM flows within the chamber, caused by the temperature gradient. The liquid PCM flow contributes to the convective heat transfer within PCM, resulting in an increase in the total heat transfer rate. In order to take the convective effect into account, the heat transfer rate of PCM should be enhanced when the PCM is melting. Moreover, when PCM is fully melted the temperature gradient between each PCM layer would continue to drive the convective heat transfer. Therefore, two assumptions are made to include the above convective heat transfer effect: (1) the heat transfer rate within PCM chamber during melting is positively related to the liquid fraction by using a logistic function; (2) the heat transfer rate between completely molten PCM nodes is positively related to their temperature gradient. Based on the assumptions, the heat transfer rate k_{PCM} inside the PCM can be expressed as:

$$k_{PCM} = \begin{cases} k_{solidus} & \text{if } T < T_s \\ k_{solidus} + \frac{k_{liquidus}}{1 + e^{\frac{\xi(T - T_s)}{T_l - T_s}}} & \text{if } T_s < T < T_l \\ (k_{solidus} + k_{liquidus})(T_n - T_{n+1})^\omega & \text{if } T \geq T_l \end{cases} \quad (17)$$

where $k_{solidus}$ is the heat conductivity of solid PCM (W/(m K)); $k_{liquidus}$ is the heat transfer rate rise at the moment when the PCM node is completely transferred into liquid state (W/(m K)); ξ and ω are constants determined by the property of PCM; T_n and T_{n+1} is the temperature of two adjacent PCM layers (°C).

Based on Eq. (17), the profile of RT35 heat transfer rate during its melting process is simulated and shown in Fig. 6.

To calibrate the thermal resistance model, meaning determining the value of two constants ξ and ω in Eq. (17), the result from CFD simulation is used to compare with the result from the proposed new model as the CFD simulation method is widely used in the research of PV-PCM system. The value of $k_{liquidus}$ is the averaged heat transfer rate rise derived from CFD model at the moment when all PCM grids are completely molten and is determined as 4.82. To determine the value of ξ

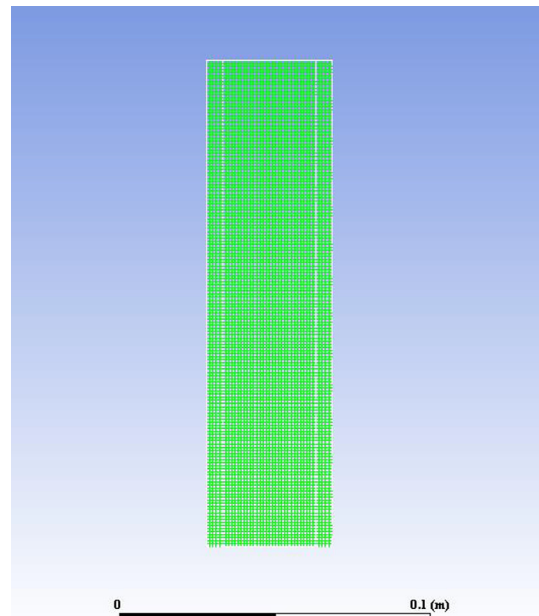


Fig. 2. A sample mesh of the PV-PCM system.

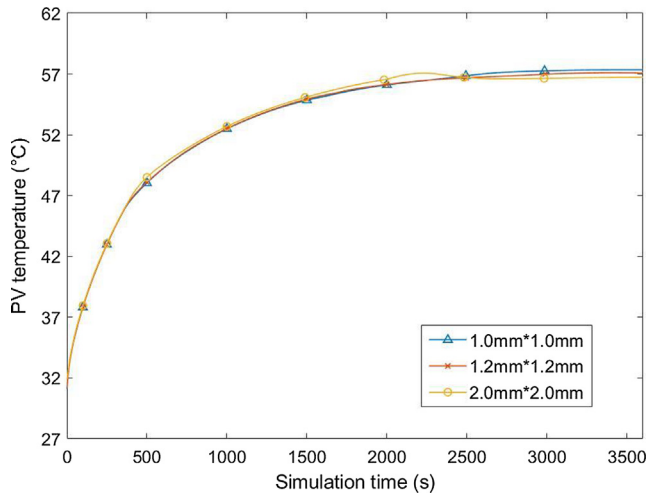


Fig. 3. The grid sensitivity test of the PV-PCM system with different cell sizes.

and ω , melting processes and convection processes simulated by CFD model were compared with results of 1-D model according to the averaged discrepancy caused by varying these two values. Finally, ξ and ω is determined as 10 and 2 respectively. After calibration, a PV-PCM system, with 30 mm RT35 PCM under 27 °C ambient temperature and 900 W/m² solar radiation, was simulated for validation. To be noted that in simulation, the PCM chamber was not finned due to the difficulties in incorporating fins into 1-D thermal resistance model and the effect of fins would be equivalent to certain changes in thermal conductivity. Finally, according to the simulations, the root-mean-square deviation (RMSD) between two methods is 0.173 and the normalized RMSD calculated by the range of data is 0.5%, presenting a high degree of agreement as shown in Fig. 7. Moreover, the improved 1-D model shows great advantage in shortening computation time, i.e., over 300 times faster than the CFD model in this case. Therefore, it is indicated that the proposed new model is reliable and feasible for further investigation.

4. Study on different assumptions in mathematical modelling

In the literature, various assumptions have been made in their mathematical models but the impact of those assumptions on result has not been sufficiently reported. Therefore, as a first attempt, the significance of different assumptions on simulation result is examined to ensure its accuracy for the further analysis. Three commonly used assumptions are summarized as follows:

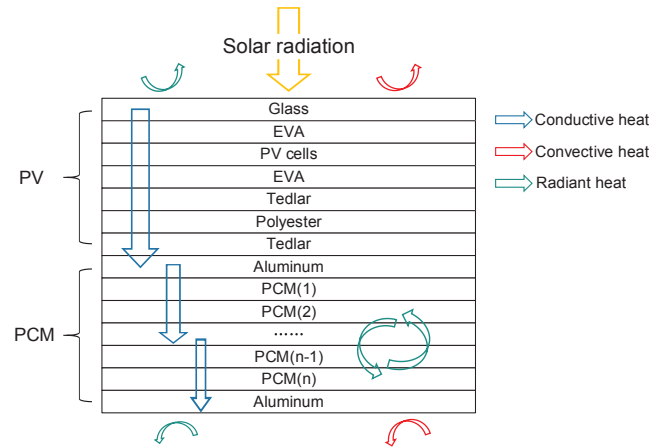


Fig. 5. Schematic diagram of 1-D thermal resistance model.

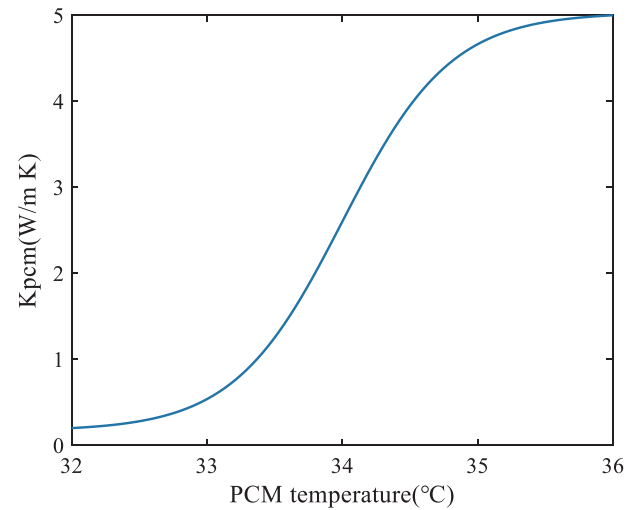


Fig. 6. The heat transfer rate variation during the melting process.

- Assumption 1: only conductive heat transfer is considered within PCM and the convective heat transfer within liquid PCM is neglected for simplification [41,43];
- Assumption 2: only natural air convection or forced air convection is considered at the surrounding of system, while the radiative heat transfer is neglected [56,57];
- Assumption 3: The PV module is treated as a layer of aluminum plate for simplification [31].

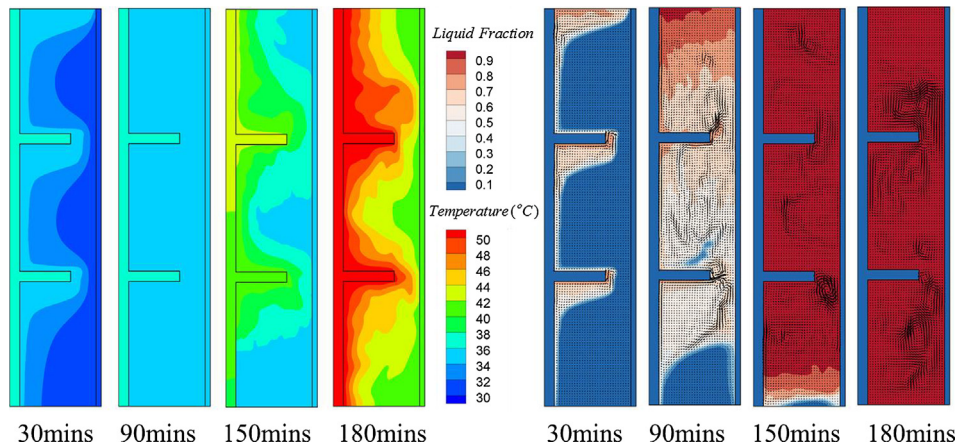


Fig. 4. The CFD simulation result of a PV-PCM system.

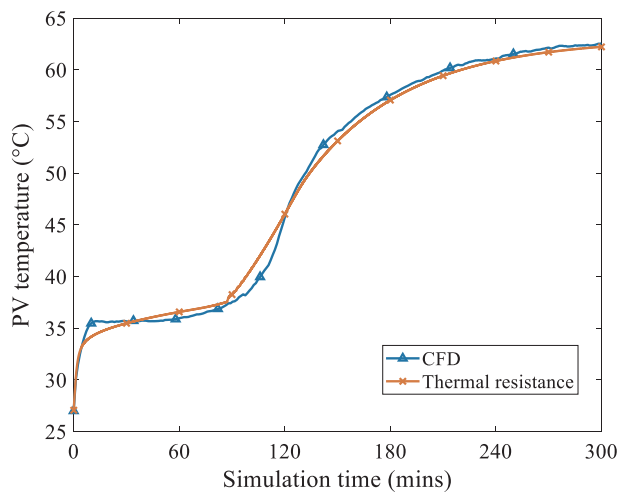


Fig. 7. PV temperature profile modelled by CFD method (FLUENT program) and thermal resistance method (MATLAB program).

To examine the above assumptions, a typical PV-PCM system is simulated under ambient temperature at 25 °C, solar radiation at 800 W/m², thickness of RT35 PCM at 30 mm, and wind speed at zero.

4.1. Assumption 1 – neglect of convective heat transfer within liquid PCM

As mentioned earlier, due to the inherent weakness of 1-D thermal resistance model, the convective heat transfer in liquid PCM is usually neglected and only conductive heat transfer is considered. The convective effect, however, could be significant as the temperature gradient induces considerable flow in liquid PCM when the PCM is melting, and the heat transfer rate is thus enhanced. Based on the newly developed model for PCM heat transfer rate in Eq. (17), the impact of such convective heat transfer on the PV temperature can be simulated.

As presented in Fig. 8, a significant gap between the two temperature profiles can be observed. When the convective heat transfer in PCM is considered, the PV temperature is 10 °C lower than the one excluded. Therefore, the conclusion is that Assumption 1 can lead to some significant error and the convective heat transfer should be considered in the simulation.

4.2. Assumption 2 – neglect of radiative heat transfer with ambient

It is worthwhile to examine the impact of neglect of radiative heat transfer with ambient on system performance. The result in Fig. 9 demonstrates that, if the radiative heat dissipation with the sky and the ground is not considered, the melting time of PCM in the system is reduced and the PV temperature increases more quickly. At the end of simulation, the temperature difference can be up to about 20 °C as the heat radiation plays a very important role in heat transfer under high temperature. Thus, it can be concluded that radiative heat transfer should be included.

4.3. Assumption 3 – simplification of PV panel

The PV module is simplified as an aluminum plate in Ref. [31]. However, as the PV module is usually composed of glass, PV cell, Ethylene-vinyl acetate (EVA) and Tedlar-Polyester-Tedlar (TPT), the simplification of PV module as an aluminum plate could lead to a quite different specific heat capacity and conductive heat transfer rate. The effect of such simplification is investigated in this section. The result in Fig. 10 illustrates that the difference between two temperature profiles is not very significant, with an average difference at about ± 1.5 °C. Therefore, the conclusion from this section is that the simplification of PV module as an aluminum plate can bring some errors in result while it

might be acceptable if some deviation is allowed.

5. Sensitivity analysis of the PV-PCM system design

After taking both convective and radiative effect into account, the improved thermal resistance model is finally adopted for optimizing PV-PCM system design from over 300 cases with different configurations and different conditions, i.e. ambient temperature, wind speed and solar radiation.

5.1. Investigation of the melting process

To understand heat transfer process and melting process in the system, a specific case was modeled as an attempt. The simulation result is presented in Fig. 11 by using the contour map. It takes more time to increase PV temperature from 35 °C to 40 °C in contrast to other temperature intervals. This is reasonable because the melting temperature range of RT35 is 29–36 °C, indicating that during the phase change period, huge amount of latent heat can be absorbed and stored in PCM. In total it takes about 120 min to melt the PCM completely, and then the PV surface temperature rises rapidly from 40 °C to over 60 °C in a short time. Specifically, it takes only about 15 min to increase from 40 °C to 45 °C while it needs about 100 min from 35 °C to 40 °C. This phenomenon demonstrates the role of the PCM in PV temperature control, particularly around the phase change temperature.

The melting process of PCM can be observed in Fig. 11(b), which presents the liquid fraction along the PCM depth during the melting process. It is worthy to point out that although it causes exactly same amount of energy to melt every 10% of PCM, it takes less time to melt the first several tenth of PCM than others afterward. This could be explained that the heat removal caused by natural air convection becomes intensive as the PV temperature rises. In this example, 143 min are required for PCM melting completely.

5.2. Sensitivity analysis on a single variable

In this section, the effect of changing several key parameters on simulation result, i.e. ambient temperature, solar radiation, wind speed, PCM thickness and latent heat capacity, has been investigated. Specifically, results of 300 cases are graphically illustrated from Sections 5.2–5.3. Configurations and boundary conditions of the cases are also described in these sections. Such sensitivity analysis can help the designers understand the effect of uncertainty, and it can also help the modelers to determine the impact that variations in assumed inputs have on the behavior, feasibility, and economics of a particular system

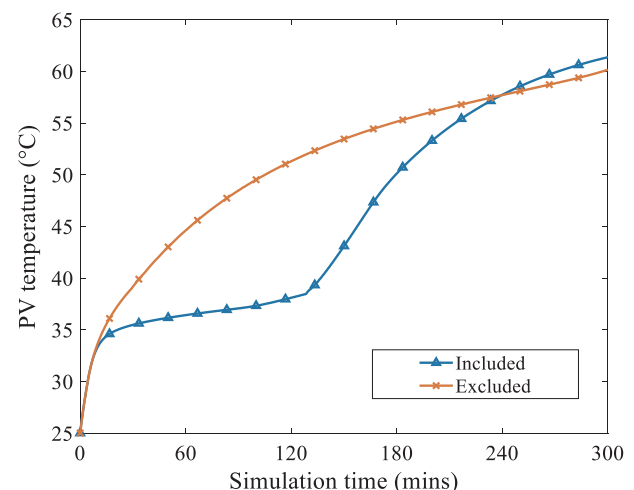


Fig. 8. PV temperature profile of PV-PCM system when the convective heat transfer is included and excluded.

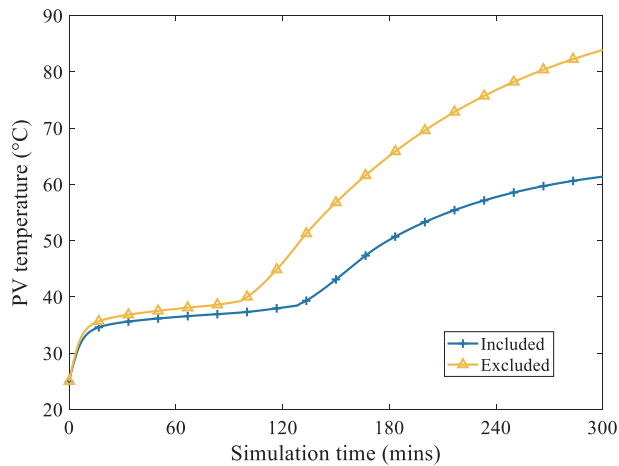


Fig. 9. PV temperature profile of PV-PCM system when the radiative heat transfer is included and excluded.

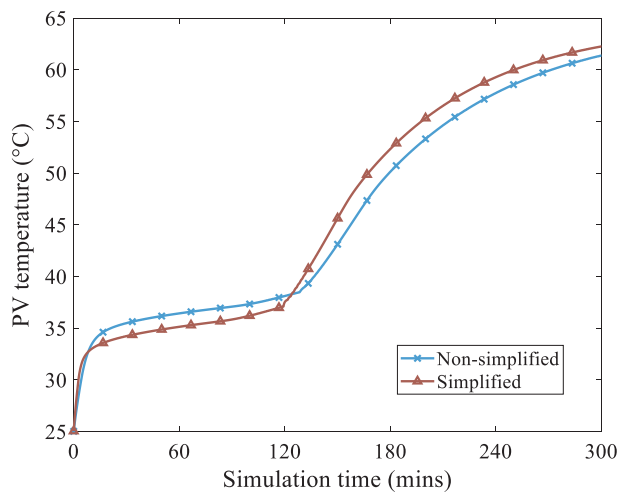


Fig. 10. PV temperature profile of PV-PCM system when the PV module is simplified as an aluminum plate and non-simplified.

configuration. Sensitivity analysis can also check the robustness of a particular system configuration. That is whether it is optimal in other scenarios when initial conditions have been changed.

5.2.1. Ambient temperature

A sensitivity analysis on ambient temperature can help users to understand the performance of a particular system in different locations or seasons. Fig. 12 provides that as the ambient temperature increases, the simulated PV temperature increases as well and the melting time is reduced. The total PCM melting process only takes about 1 h for ambient temperature at 30 °C, while it takes more than 2 h when ambient temperature is 20 °C.

To achieve a better performance of PV-PCM system under high ambient temperature condition, like Shanghai, a high melting temperature or high latent heat capacity is recommended. Fig. 12 also suggests that 5 °C increase in ambient temperature can lead to almost the same increase in the PV temperature at the end of simulation.

5.2.2. Solar radiation intensity

This section examines how the solar radiation intensity affects the system performance. As shown in Fig. 13, solar radiation has significant impact on the system performance, a higher solar radiation results in a shorter melting period and consequently a higher PV temperature. At the end, all PCM is molten in these cases. On average, a 100 W/m²

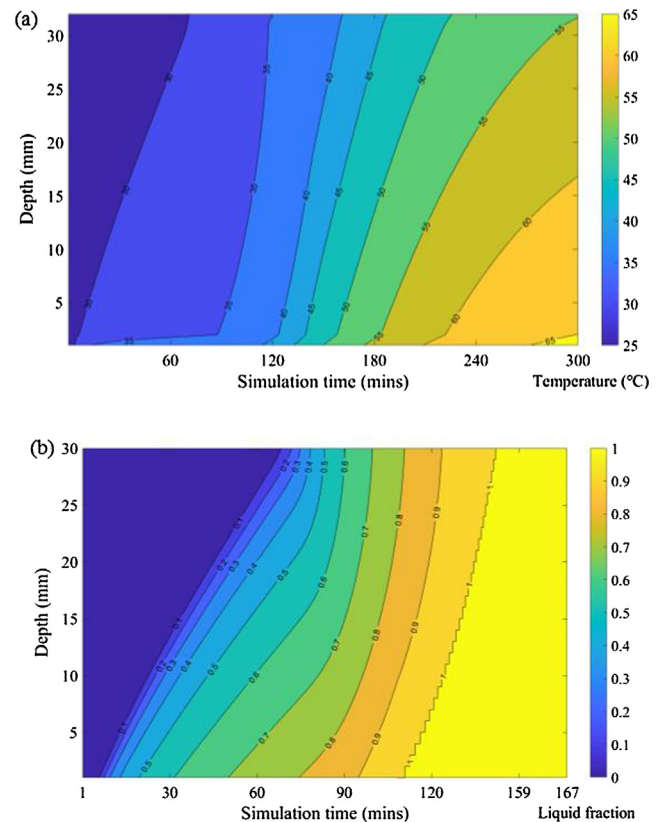


Fig. 11. (a) PV temperature variation and (b) liquid fraction variation at different depth of PV-PCM system during the melting process (PCM: RT35, PCM thickness: 30 mm, ambient temperature: 25 °C, solar radiation: 900 W/m², wind speed: 0 m/s).

increase in solar radiation can result in a 5 °C increase in PV temperature when PCM is totally molten and PV temperature becomes stable. This relation could be applied to predict temperature variance in real applications.

5.2.3. Wind speed

In this section, the effect of surrounding wind speed on system performance is discussed, this has not been sufficiently examined by other studies according to the knowledge of the authors. Fig. 14 illustrates that the heat dissipation effect of wind is significant when the

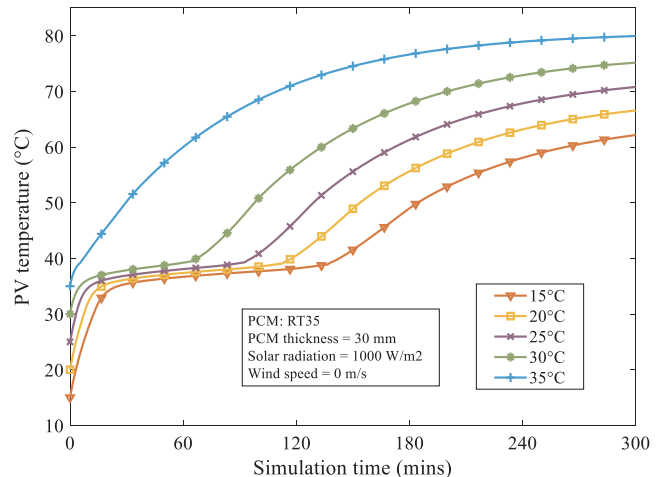


Fig. 12. PV temperature profile of PV-PCM system under different ambient temperature.

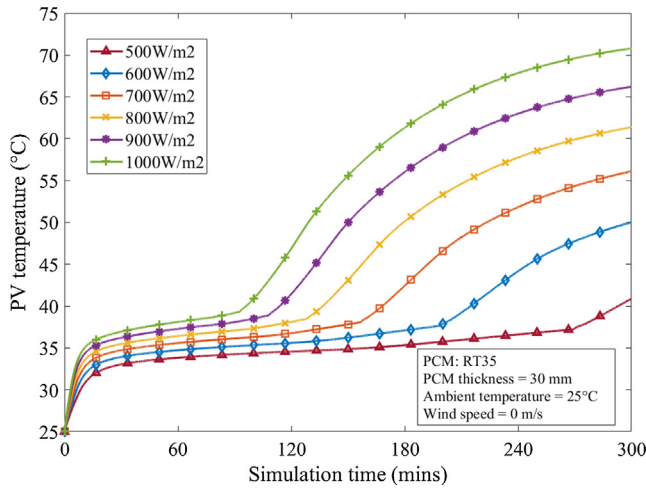


Fig. 13. PV temperature profile of PV-PCM system under different solar radiation intensity.

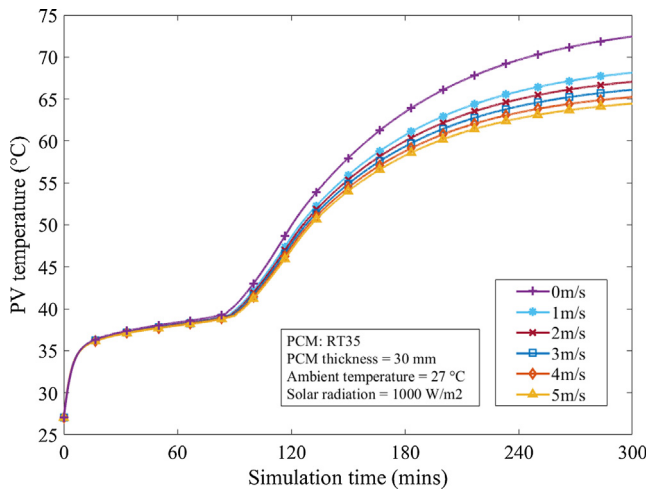


Fig. 14. PV temperature profile of PV-PCM system under different wind speed.

wind speed increases from 0 m/s to 1 m/s, with PV temperature reduced by about 5 °C. However, such temperature reduction is not so obvious when the wind speed continues to rise. Fig. 14 also suggests that different wind speed has little impact on PCM melting time and PV temperature.

5.2.4. PCM thickness

Fig. 15 investigates the effect of the thickness of PCM, one of the key parameters of PV-PCM system. The result shows that PV temperature in PV-PCM system could be reduced by up to 35 °C when compared to traditional PV-only module. Correspondingly, the maximum output gap between two systems can be enlarged to 17.5% of PV reference output power. It is observed in Fig. 15 that every 5 mm increase in thickness prolongs the melting period by about 10 min on average.

Another finding is that when the PCM is fully molten in the end, the PV temperature of PV-PCM system could be higher than that of traditional PV-only system. Thus, sometimes PCM could become a thermal resistance for heat dissipation at the back of PV panel, increasing PV temperature conversely. This phenomenon might also happen in winter when temperature of PCM would not reach the melting point. It highlights that a proper selection of PCM thickness and melting point according to local climate is really important. Therefore, further research is suggested to deal with the real climate condition for a whole year.

5.2.5. PCM latent heat capacity

Another impact factor is the latent heat capacity of PCM. Fig. 16 presents a sensitivity analysis on latent heat capacity of PCM RT35. The effect of latent heat capacity on simulation results is quite uniform, every 40 kJ/kg increase in latent heat capacity results in about 20 min' increase in melting time. Similar to Fig. 15, Fig. 16 also indicates that the molten PCM could become a thermal resistance.

5.3. Sensitivity analysis on two variables

A two-variable analysis method has been adopted in this study to optimize system configuration. The improved thermal resistance model is employed to simulate different system configurations by changing two key parameters simultaneously, including PCM thickness, PCM types, solar radiation, ambient temperature and wind speed. After simulation, a 2-D profile of PV temperature and power generation, which is also called contour map, based on two variables can be obtained. The two-variable sensitivity analysis method can demonstrate the system performance vividly, which is very helpful for researchers and engineers to optimize a system design if two variables should be considered simultaneously.

5.3.1. Effects of solar radiation and PCM thickness

Figs. 17 and 18 present the system performance simulated at the ambient temperature of 25 °C. RT35 is used as PCM in this analysis. Based on Fig. 17, the optimal PCM thickness can be determined according to solar radiation intensity and the targeted PV temperature. For example, if PV temperature should be controlled under 60 °C when solar radiation is 800 W/m², the thickness should be at least 35 mm. Besides, according to the slope of isothermal line in Fig. 17, it would be very difficult to control the PV temperature to a low value if only PCM thickness is increased when solar radiation is at a high level. However, it does not mean that it is infeasible to apply PV-PCM system in high radiation area.

Fig. 18 presents the electricity production enhancement of PV-PCM system compared to traditional PV-only module under the same condition. Indeed, it reveals that the PV-PCM system can perform better as the solar radiation rises. The latent heat capacity of PCM can control PV temperature around its melting temperature while the temperature of PV-only system rises rapidly at high solar radiation level. Therefore, the temperature difference between two systems is significant, resulting in a substantial increase in PV conversion efficiency when the PCM is added. As a whole, the improvement ranges from 5% to 12% after 5-h simulation.

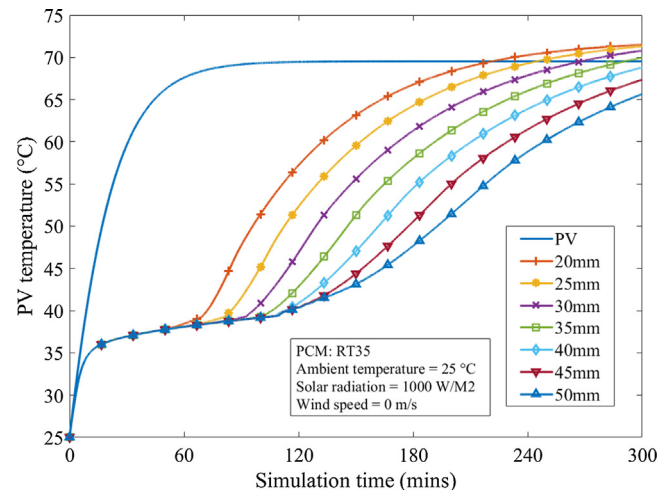


Fig. 15. PV temperature profile of PV-PCM systems with different PCM thickness.

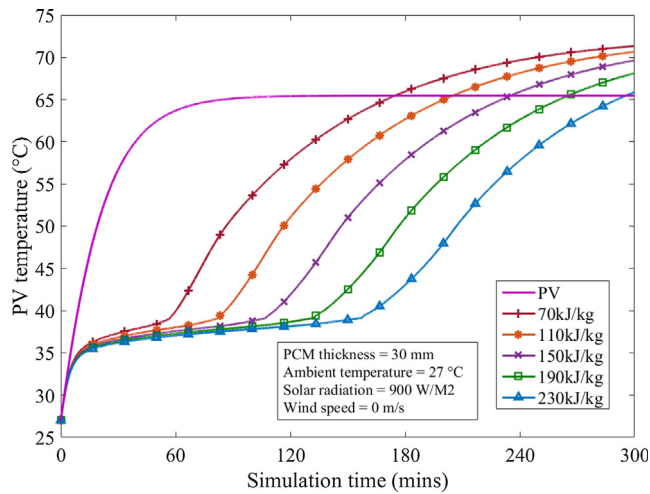


Fig. 16. PV temperature profile of PV-PCM systems with different latent heat capacities.

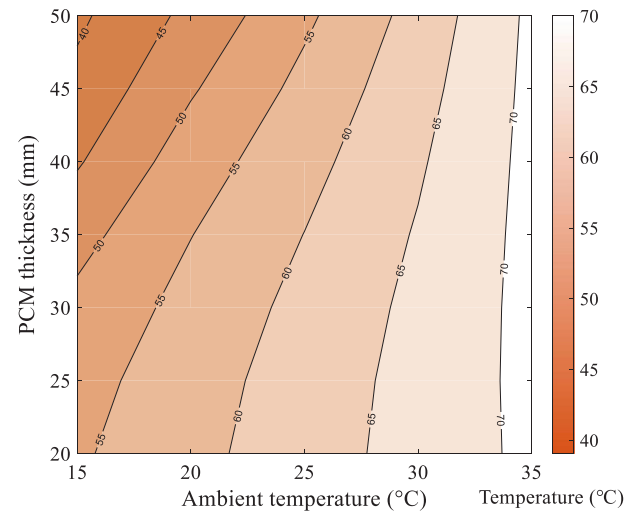


Fig. 19. PV temperature of systems with different PCM thickness and ambient temperature after five-hour simulation.

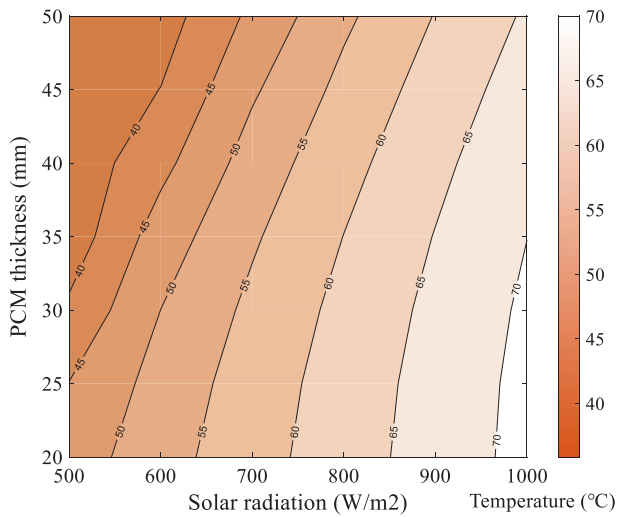


Fig. 17. PV temperature of systems with different PCM thickness and solar radiation after 5 h simulation.

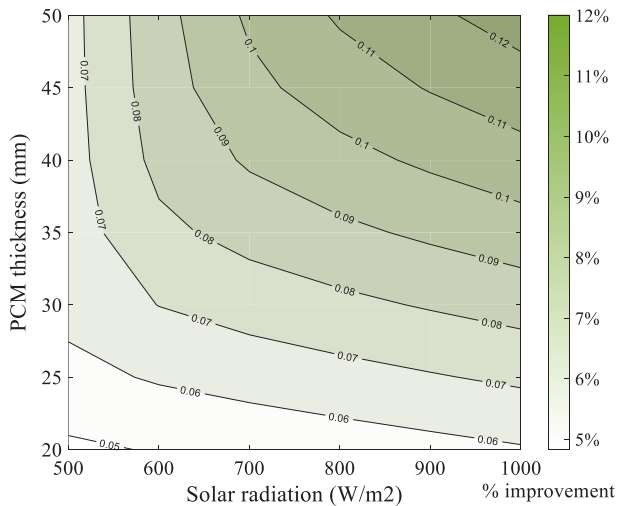


Fig. 18. The electricity production improvement of PV-PCM system with different PCM thickness and solar radiation after 5 h simulation.

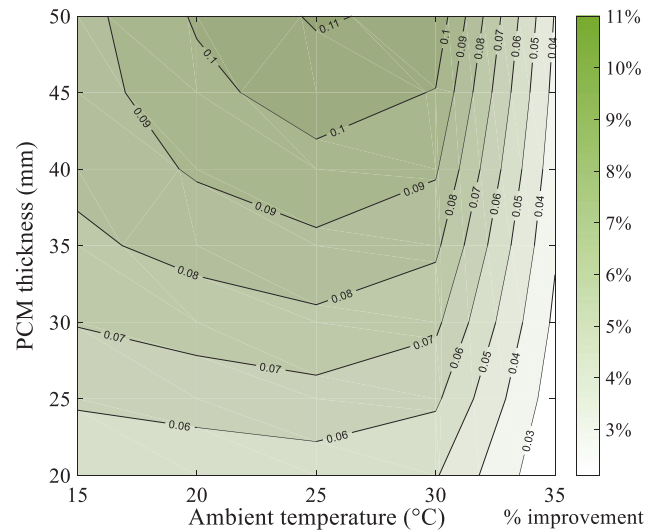


Fig. 20. The electricity production improvement of PV-PCM systems with different PCM thickness and ambient temperature after 5-h simulation.

5.3.2. Effects of ambient temperature and PCM thickness

Figs. 19 and 20 discuss the effect of PCM thickness and ambient temperature. PCM RT35 is used and the solar radiation is set as 800 W/m². Fig. 19 shows that if the ambient temperature is higher than the melting temperature range of PCM, i.e. 29–36 °C, an increase in PCM thickness has no effects in PV temperature control. As illustrated in Fig. 20, when ambient temperature changes from 15 to 35 °C, the PV power production improvement increases in the beginning, peaking at ambient temperature of about 25 °C and then followed by an obvious descending tendency. This interesting phenomenon could be explained by the temperature gap between PV-only system and PV-PCM system during the 5-h simulation. In the beginning when ambient temperature rises from 15 to 25 °C, the latent heat storage capacity of PCM can successfully control the PV temperature at a lower value than the PV-only system, therefore, the PV power improvement is significant in this range. However, if ambient temperature continues to rise from 25 to 35 °C, the melting time of PCM has been shortened greatly due to reduced latent heat capacity since the ambient temperature is close to or over the low boundary of melting temperature range, which makes the PV temperature in PV-PCM system has a close or equal value with PV-only system. Therefore, the power enhancement effect of PCM is not obvious

at high ambient temperatures. In this regard, as demonstrated in this case, it is recommended to use PCM in real applications with slightly higher melting temperature than the ambient temperature, for example 5 °C, in order to maximize the enhancement in PV conversion efficiency.

6. Conclusions

In this study, the mathematical model of the PV-PCM system is developed. The CFD model and 1-D thermal resistance model are compared for numerical study on the proposed PV-PCM system. It was found that the CFD simulation cannot perform large number of simulation cases while the traditional 1-D thermal resistance model cannot solve the momentum equations of the molten PCM. Therefore, an improved thermal resistant model is developed, which consumes less computation time than CFD method and can incorporate the convective heat transfer effect within melted PCM through applying the enhanced conductivity method. The newly developed equation for convective heat transfer was simulated in MATLAB software and validated through CFD method, presenting a high degree of agreement. The improved thermal resistance model was then further employed to examine the impacts of several common assumptions in mathematical modelling. The result shows that the neglect of convective heat transfer, neglect of radiative heat transfer, and simplification of PV module as an aluminate plate could result in about 20, 10 and ± 1.5 °C difference in PV temperature respectively, indicating that the first two assumptions are unreasonable. Moreover, a study of over 300 cases with different configurations has been completed. It is observed that solar radiation is a dominant factor for determining the PV temperature. The results illustrate that, on average, every 100 W/m² increase in solar radiation can lead to 5 °C increase in PV temperature. Furthermore, a two-variable sensitivity analysis is performed, illustrating that the PV-PCM system has great potential for implementation in high solar radiation area, and around 5 °C higher than the ambient temperature is recommended for the PCM's phase change temperature. It is believed that the findings from this study can provide some reference for optimizing PV-PCM system in real application. By integrating the two methods, i.e., the enhanced conductivity method and two-variable sensitivity analysis, future study would be carried out in simulation under real climate conditions during a year-round time period. The enhanced conductivity method can also be further developed by using dimensionless parameters to expand its application in different geometries and properties.

Acknowledgments

The authors would appreciate the financial supports provided by the National Natural Science Foundation of China (NSFC) through the Grant 51506183. Special thanks would be given to the colleagues from Renewable Energy Research Group in The Hong Kong Polytechnic University who provided great support for this project.

References

- Ma T, Yang H, Lu L. A feasibility study of a stand-alone hybrid solar–wind–battery system for a remote island. *Appl Energy* 2014;121:149–58.
- National Renewable Energy Laboratory, Best Research-Cell Efficiencies, Retrieved from: < <https://www.nrel.gov/pv/assets/images/efficiency-chart.png> >; 2018.
- Ma T, Yang H, Lu L. Solar photovoltaic system modeling and performance prediction. *Renew Sustain Energy Rev* 2014;36:304–15.
- Singh GK. Solar power generation by PV (photovoltaic) technology: a review. *Energy*. 2013;53:1–13.
- Ma T, Yang H, Lu L. Development of a model to simulate the performance characteristics of crystalline silicon photovoltaic modules/strings/arrays. *Sol Energy* 2014;100:31–41.
- Tian Y, Zhao CY. A review of solar collectors and thermal energy storage in solar thermal applications. *Appl Energy* 2013;104:538–53.
- Skoplaki E, Palyvos JA. On the temperature dependence of photovoltaic module electrical performance: a review of efficiency/power correlations. *Sol Energy* 2009;83:614–24.
- Ma T, Yang H, Lu L. Performance evaluation of a stand-alone photovoltaic system on an isolated island in Hong Kong. *Appl Energy* 2013;112:663–72.
- van Helden WJ, van Zolingen RJC, Zondag HA. PV thermal systems: PV panels supplying renewable electricity and heat. *Prog Photovolt Res Appl* 2004;12:415–26.
- Ma T, Yang H, Zhang Y, Lu L, Wang X. Using phase change materials in photovoltaic systems for thermal regulation and electrical efficiency improvement: a review and outlook. *Renew Sustain Energy Rev* 2015;43:1273–84.
- Brahim T, Jemni A. Economical assessment and applications of photovoltaic/thermal hybrid solar technology: a review. *Sol Energy* 2017;153:540–61.
- Siecker J, Kusakana K, Numbi BP. A review of solar photovoltaic systems cooling technologies. *Renew Sustain Energy Rev* 2017;79:192–203.
- Du D, Darkwa J, Kokogiannakis G. Thermal management systems for Photovoltaics (PV) installations: a critical review. *Sol Energy* 2013;97:238–54.
- Hasan A, McCormack S, Huang M, Norton B. Evaluation of phase change materials for thermal regulation enhancement of building integrated photovoltaics. *Sol Energy* 2010;84:1601–12.
- Stultz JWWL. Thermal performance testing and analysis of photovoltaic modules in natural sunlight. Pasadena (California): Jet Propulsion Laboratory; 1978.
- Huang M, Eames P, Norton B. Phase change materials for limiting temperature rise in building integrated photovoltaics. *Sol Energy* 2006;80:1121–30.
- Huang MJ, Eames PC, Norton B. Thermal regulation of building-integrated photovoltaics using phase change materials. *Int J Heat Mass Transf* 2004;47:2715–33.
- Huang M, Eames P, Norton B, Hewitt N. Natural convection in an internally finned phase change material heat sink for the thermal management of photovoltaics. *Sol Energy Mater Sol Cells* 2011;95:1598–603.
- Huang MJ, Eames PC, Norton B. Comparison of predictions made using a new 3D phase change material thermal control model with experimental measurements and predictions made using a validated 2D model. *Heat Transf Eng* 2007;28:31–7.
- Hasan A, McCormack SJ, Huang MJ, Norton B. Energy and cost saving of a photovoltaic-phase change materials (PV-PCM) system through temperature regulation and performance enhancement of photovoltaics. *Energies* 2014;7:1318–31.
- Hasan A, Sarwar J, Alnoman H, Abdelbaqi S. Yearly energy performance of a photovoltaic-phase change material (PV-PCM) system in hot climate. *Sol Energy* 2017;146:417–29.
- Hasan A, Alnoman H, Rashid Y. Impact of integrated photovoltaic-phase change material system on building energy efficiency in hot climate. *Energy Build* 2016;130:495–505.
- Elarga H, Goia F, Zarrella A, Dal Monte A, Benini E. Thermal and electrical performance of an integrated PV-PCM system in double skin façades: a numerical study. *Sol Energy* 2016;136:112–24.
- Park J, Kim T, Leigh S-B. Application of a phase-change material to improve the electrical performance of vertical-building-added photovoltaics considering the annual weather conditions. *Sol Energy* 2014;105:561–74.
- Japs E, Sonnenrein G, Krauter S, Vrabec J. Experimental study of phase change materials for photovoltaic modules: energy performance and economic yield for the EPEX spot market. *Sol Energy* 2016;140:51–9.
- Karthick A, Murugavel KK, Ramanan P. Performance enhancement of a building-integrated photovoltaic module using phase change material. *Energy* 2017.
- Smith CJ, Forster PM, Crook R. Global analysis of photovoltaic energy output enhanced by phase change material cooling. *Appl Energy* 2014;126:21–8.
- Koşny J, Biswas K, Miller W, Kriner S. Field thermal performance of naturally ventilated solar roof with PCM heat sink. *Sol Energy* 2012;86:2504–14.
- Khanna S, Reddy K, Mallick TK. Performance analysis of tilted photovoltaic system integrated with phase change material under varying operating conditions. *Energy* 2017.
- Aelenei L, Pereira R, Gonçalves H, Athienitis A. Thermal performance of a hybrid BIPV-PCM: modeling, design and experimental investigation. *Energy Procedia* 2014;48:474–83.
- Biwole PH, Eclache P, Kuznik F. Phase-change materials to improve solar panels performance. *Energy Build* 2013;62:59–67.
- Maiti S, Banerjee S, Vyas K, Patel P, Ghosh PK. Self regulation of photovoltaic module temperature in V-trough using a metal–wax composite phase change matrix. *Sol Energy* 2011;85:1805–16.
- Emam M, Ookawara S, Ahmed M. Performance study and analysis of an inclined concentrated photovoltaic-phase change material system. *Sol Energy* 2017;150:229–45.
- Rea JE, Oshman CJ, Olsen ML, Hardin CL, Glatzmaier GC, Siegel NP, et al. Performance modeling and techno-economic analysis of a modular concentrated solar power tower with latent heat storage. *Appl Energy* 2018;217:143–52.
- Klugmann-Radziemska E, Weislo-Kucharek P. Photovoltaic module temperature stabilization with the use of phase change materials. *Sol Energy* 2017;150:538–45.
- Malvi C, Dixon-Hardy D, Crook R. Energy balance model of combined photovoltaic solar-thermal system incorporating phase change material. *Sol Energy* 2011;85:1440–6.
- Cui T, Xuan Y, Li Q. Design of a novel concentrating photovoltaic–thermoelectric system incorporated with phase change materials. *Energy Convers Manage* 2016;112:49–60.
- Sharma S, Micheli L, Chang W, Tahir AA, Reddy KS, Mallick TK. Nano-enhanced phase change material for thermal management of BICPV. *Appl Energy* 2017;208:719–33.
- Ho C, Tanuwijaya A, Lai C-M. Thermal and electrical performance of a BIPV integrated with a microencapsulated phase change material layer. *Energy Build* 2012;50:331–8.
- Nižetić S, Arici M, Bilgin F, Grubišić-Čabro F. Investigation of pork fat as potential novel phase change material for passive cooling applications in photovoltaics. *J*

- Clean Prod 2018;170:1006.
- [41] Atkin P, Farid MM. Improving the efficiency of photovoltaic cells using PCM infused graphite and aluminium fins. *Sol Energy* 2015;114:217–28.
 - [42] Luo Z, Huang Z, Xie N, Gao X, Xu T, Fang Y, et al. Numerical and experimental study on temperature control of solar panels with form-stable paraffin/expanded graphite composite PCM. *Energy Convers Manage* 2017;149:416–23.
 - [43] Hendricks J, Sark W. Annual performance enhancement of building integrated photovoltaic modules by applying phase change materials. *Prog Photovolt Res Appl* 2013;21:620–30.
 - [44] Kant K, Shukla A, Sharma A, Biwole PH. Heat transfer studies of photovoltaic panel coupled with phase change material. *Sol Energy* 2016;140:151–61.
 - [45] Brano VL, Ciulla G, Piacentino A, Cardona F. Finite difference thermal model of a latent heat storage system coupled with a photovoltaic device: description and experimental validation. *Renew Energy* 2014;68:181–93.
 - [46] Zhang P, Xiao X, Ma ZW. A review of the composite phase change materials: fabrication, characterization, mathematical modeling and application to performance enhancement. *Appl Energy* 2016;165:472–510.
 - [47] Jany P, Bejan A. Scaling theory of melting with natural convection in an enclosure. *Int J Heat Mass Transf* 1988;31:1221–35.
 - [48] Kahraman R, Zughbi HD, Al-Nassar YN, Hastaoglu MA, Sobh N. A simplified numerical model for melting of ice with natural convection. *Int Commun Heat Mass Transf* 1998;25:359–68.
 - [49] Holman JP. Heat transfer. New York: McGraw-Hill Education; 2009.
 - [50] Costa M, Buddhi D, Oliva A. Numerical simulation of a latent heat thermal energy storage system with enhanced heat conduction. *Energy Convers Manage* 1998;39:319–30.
 - [51] Souayfane F, Biwole PH, Fardoun F. Melting of a phase change material in presence of natural convection and radiation: a simplified model. *Appl Therm Eng* 2018;130:660–71.
 - [52] Vidalain G, Gosselin L, Lacroix M. An enhanced thermal conduction model for the prediction of convection dominated solid–liquid phase change. *Int J Heat Mass Transf* 2009;52:1753–60.
 - [53] Biwole P, Woloszyn M, Pompeo C. Heat transfers in a double-skin roof ventilated by natural convection in summer time. *Energy Build* 2008;40:1487–97.
 - [54] Bergman TL, Incropera FP. Fundamentals of heat and mass transfer. John Wiley & Sons; 2011.
 - [55] Ma T, Zhao J, Han J. A parametric study about the potential to integrate phase change material into photovoltaic panel. *Energy Procedia* 2017;142:648–54.
 - [56] Ciulla G, Brano VL, Cellura M, Franzitta V, Milone D. A finite difference model of a PV-PCM system. *Energy Procedia* 2012;30:198–206.
 - [57] Lo Brano V, Ciulla G, Piacentino A, Cardona F. On the efficacy of PCM to shave peak temperature of crystalline photovoltaic panels: an FDM model and field validation. *Energies* 2013;6:6188–210.